

SMOOTHING CALABI–YAU THREEFOLDS IN GORENSTEIN TORIC FANO FOURFOLDS

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ABSTRACT. Let $\Delta \subset \mathbb{Z}^4$ be a reflexive polytope with unit edges and let $X \subset \mathbb{P}_\Delta$ be a generic anticanonical hypersurface: a Calabi–Yau threefold whose singular locus consists of $\ell(F^\circ)$ points of type $C(F)$ for each non-smooth two-dimensional face F , where $\ell(F^\circ)$ is the lattice length of the dual edge. In a companion paper we showed that X admits no smoothing when some $C(F)$ has no smoothing component. Here we study the complementary case, in which every singular germ is locally smoothable and the question is global. We prove that the points over a face F of type (S) are smoothed by an explicit binomial deformation of the ambient toric pair whenever $\ell(F^\circ) \geq 2$, by transplanting Petracci’s homogeneous deformations of toric pairs to codimension-two faces; consequently X is smoothable whenever F is its unique non-smooth face, and unconditionally whenever every non-smooth face is a unit parallelogram with $\ell(F^\circ) \geq 2$; for several faces of general type (S) the face-wise smoothings assemble whenever the semiuniversal deformation space of X is irreducible. We show the threshold is sharp: we exhibit a reflexive polytope with 7 vertices whose hypersurface is a Calabi–Yau threefold with a single ordinary double point, which admits no smoothing by Friedman’s criterion. At $\ell(F^\circ) = 1$ the remaining germs with one interior point are the anticanonical cones over $\mathbb{P}^1 \times \mathbb{P}^1$, dP_7 and dP_6 ; we show that their toric crepant resolutions are *never* primitive contractions — the relative Picard number equals the Picard number of the del Pezzo surface — and that X is not $\mathbb{Q}$ -factorial there, so neither Friedman’s criterion nor Gross’s smoothing theorems decide these cases, which remain open. Gross’s theorem does smooth the same germs in their primitive realizations, and his exceptional surfaces $\mathbb{P}^2$ and $\mathbb{F}_1$ are precisely the two rigid classes with one interior point of the companion paper. For polytopes all of whose singular faces are unit parallelograms we compare the resulting per-face criterion with the Namikawa-criterion census of Batyrev–Kreuzer: at most one in eight of their 30,241 certified smoothings is explained face-locally, and at least 26,467 involve a cross-face homology relation.

1. INTRODUCTION

Let $\Delta \subset \mathbb{Z}^4$ be a reflexive polytope all of whose edges have lattice length one, and let $X \subset \mathbb{P}_\Delta$ be a generic anticanonical hypersurface in the Gorenstein Fano toric fourfold associated to the face fan of Δ . By Batyrev [Bat94] X is a Calabi–Yau threefold with Gorenstein canonical singularities, and by the planting proposition of the prequel [Wue26] its singular locus is exactly: for each two-dimensional face $F \preceq \Delta$ whose cone $C(F)$ is singular, $\ell(F^\circ)$ points at which the germ of X is analytically isomorphic to $C(F)$.

The deformation theory of the germs is completely understood combinatorially: by Altman [Alt97] and Corti–Filip–Petracci [CFP22], $C(F)$ is *smoothable* if and only if F admits a Minkowski decomposition into unit segments and standard triangles (type (S) in the trichotomy of [Wue26]). The prequel proved that a single face of the other two types (rigid or

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deformable-but-non-smoothable) forbids every global smoothing of X . This paper treats the remaining case: *every* singular two-dimensional face is of type (S), so every singular point of X admits a local smoothing, and the question is whether the local smoothings are realized by a global deformation. This is the classical global-defect problem: for ordinary double points it is governed by Friedman’s relation among the exceptional curve classes [Fri86, RT09], and for the toric hypersurfaces at hand it was studied, in the all-conifold case, by Batyrev–Kreuzer [BK10], who found that of the 198,849 reflexive 4-polytopes whose two-dimensional faces are standard triangles and unit parallelograms (at least one parallelogram), only 30,241 could be certified smoothable by Namikawa’s criterion. One dimension down, the analogous constructive problem — smoothing Gorenstein toric Fano *threefolds* from Minkowski data on the facets of a reflexive 3-polytope — was recently solved by Corti–Hacking–Petracci [CHP24].

Our first result is constructive and identifies a clean sufficient condition. Write $\ell(F^\circ)$ for the lattice length of the edge of Δ° dual to a two-dimensional face F .

Theorem 1.1 (= Theorem 4.1). *Let Δ be reflexive with unit edges and let F be a two-dimensional face of type (S), with a Minkowski decomposition $F = Q_0 + \cdots + Q_r$ into unit segments and standard triangles, and suppose $\ell(F^\circ) \geq 2$. Then there is an explicit r -parameter deformation of the pair $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$, given by binomial equations in Cox coordinates, whose induced deformation of the generic anticanonical X has generic fibre smooth in a neighbourhood of the $\ell(F^\circ)$ points of type $C(F)$. In particular, if F is the unique non-smooth two-dimensional face of Δ , then X is smoothable.*

For several type-(S) faces the construction clears each face individually; the face-wise smoothings assemble unconditionally in the nodal case, and in general under a hypothesis on the deformation space. (The fibres of the face family are no longer toric, so the construction cannot simply be iterated; see Remark 4.5.)

Theorem 1.2 (= Theorem 4.4). *Let Δ be reflexive with unit edges and suppose that every non-smooth two-dimensional face $F_1, \dots, F_s$ of Δ is of type (S) with $\ell(F_i^\circ) \geq 2$. Then:*

- (1) *for every singular point p of X there is an explicit deformation of X whose generic fibre is smooth in a neighbourhood of p ; consequently the locus $D_p \subset (B, 0)$ of points of the semiuniversal deformation of X whose fibres remain singular near p is a proper closed analytic subgerm;*
- (2) *if the base germ $(B, 0)$ of the semiuniversal deformation of X is irreducible — for instance smooth — then X is smoothable;*
- (3) *if every F_i is a unit parallelogram, then X is smoothable unconditionally.*

The hypothesis $\ell(F^\circ) \geq 2$ is not an artifact of the method.

Theorem 1.3 (= Theorem 5.1). *Let*

$$\Delta_N = \text{conv}(\square \times \{(1, 0)\} \cup \{e_4, (0, 1, -1, 0), (-1, -2, 1, -1)\}) \subset \mathbb{Z}^4,$$

where $\square$ is the unit square embedded with vertices $(0, 0), (0, -1), (1, -1), (1, 0)$. Then Δ_N is reflexive with 7 vertices and unit edges; its unique non-smooth two-dimensional face is $\square \times \{(1, 0)\}$, of type (S), with dual edge of lattice length 1; the generic anticanonical X_N is a Calabi–Yau threefold with exactly one ordinary double point; and X_N admits no smoothing.

Thus at $\ell(F^\circ) = 1$ the global defect is real even though the local model is the simplest smoothable singularity there is. The obstruction is Friedman’s: the maximal projective crepant partial resolution of X_N is a *projective small* resolution, so the exceptional curve class cannot vanish, while a one-node Calabi–Yau threefold is smoothable only if it does. By contrast, for germs whose crepant resolution is divisorial the defect disappears:

Proposition 1.4 (= Proposition 6.1). *Let Δ be reflexive with unit edges whose unique non-smooth two-dimensional face F has exactly one interior lattice point, is of type (S), and has $\ell(F^\circ) = 1$. Then the singular locus of X is one point, the anticanonical cone over $S \in \{\mathbb{P}^1 \times \mathbb{P}^1, \text{dP}_7, \text{dP}_6\}$, and an MPCP resolution $\hat{X} \rightarrow X$ can be chosen whose exceptional locus is the smooth irreducible surface S , contracted to the point. However, the contraction has relative Picard number $\rho(S) \in \{2, 3, 4\}$ — so it factors in the projective category and is never a primitive contraction — and X is not $\mathbb{Q}$ -factorial at the point. In particular neither Gross’s theorem on primitive type II contractions [Gro97, Thm. 5.8] nor his smoothing theorem for $\mathbb{Q}$ -factorial contractions [Gro97, Thm. 4.3] applies, and the smoothability of X is open (Question 8.1).*

The statements assemble into a germ-dependent picture of the global defect (Table 1): for type-(S) faces the dual-edge length $\ell(F^\circ) \geq 2$ clears the face, and the clearings assemble in the nodal case and, in general, whenever the deformation space of X is irreducible (Theorem 1.2); at $\ell(F^\circ) = 1$ the picture is governed by the crepant resolution over the point. For the node the resolution is small and Friedman’s relation obstructs (Theorem 1.3). For the $i = 1$ germs the resolution is divisorial, but — in contrast to the primitive type II contractions analyzed by Gross — Proposition 1.4 shows the toric contraction always has relative Picard number ≥ 2 and X is not $\mathbb{Q}$ -factorial, so the two known mechanisms (Friedman’s obstruction; Gross’s smoothing theorems) both fall silent and the case is open. At the level of germs the alignment is still pleasant: Gross’s theorem for primitive realizations fails exactly for $E \cong \mathbb{P}^2$ and $E \cong \mathbb{F}_1$, and these are precisely the two classes with one interior point that are *rigid* in the trichotomy of [Wue26]. The local trichotomy and the global defect see the same dichotomy.

germ of the type-(S) face	$\ell(F^\circ) = 1$	$\ell(F^\circ) \geq 2$
unit square (node; crepant res. small)	non-smoothable (Thm 1.3)	smoothable (Thm 1.1)
$i = 1$ (cone over $\mathbb{P}^1 \times \mathbb{P}^1, \text{dP}_7, \text{dP}_6$; divisorial)	open: never primitive torically (Prop. 1.4)	smoothable (Thm 1.1)
$i \geq 2$ type (S) (divisorial, reducible exceptional locus)	open	smoothable (Thm 1.1)

TABLE 1. The global defect by germ and dual-edge length, for a single singular face. For several singular faces see Theorem 1.2: each face is cleared individually, the all-conifold case is unconditional, and the general case assembles when the deformation space is irreducible; mixed configurations involving $\ell = 1$ nodes are governed by cross-face Friedman relations (§7).

Section 7 confronts this picture with the Batyrev–Kreuzer census. For the $\ell(F^\circ)$ nodes over a single square face the exceptional classes coincide, so Friedman’s all-nonzero relation is satisfiable face-locally if and only if $\ell(F^\circ) \geq 2$ — reproducing the threshold of Theorem 1.1 from pure homology. A scan of the entire Kreuzer–Skarke list (§7) shows that this per-face condition accounts for at most 3,774 of the 30,241 conifold polytopes that Batyrev–Kreuzer certify smoothable — about one in eight — so the great majority of their smoothings are *cross-face* rescues.

All computations are performed by the exact-arithmetic programs of the prequel, extended by a scanner for conifold configurations; they are included as ancillary files and are public, with all data, at

<https://github.com/bwuebben/calabi-yau-smoothability>.

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2. PRELIMINARIES

We keep the notation of [Wue26], which the reader should consult for details; we recall what is used. For a lattice polygon Q with unit edges placed at height one, $C(Q)$ denotes the associated Gorenstein toric affine threefold. Its reduced miniversal base has one smooth component per maximal Minkowski decomposition of Q [Alt97], the smoothing components corresponding to decompositions into unit segments and standard triangles [CFP22]; Q is of *type (S)* when such a decomposition exists. The *cascade lemma* [Wue26, Lemma 2.4] describes the general fibre of Altmann's homogeneous deformation attached to a decomposition $Q = R_0 + \cdots + R_r$, realized on the Cayley cone $\text{Cone}(\bigcup_i R_i \times \{f_i\})$: it has exactly one singular point of type $C(R_i)$ for each two-dimensional non-smooth summand, and no others; in particular the general fibre is *smooth* when all summands are unit segments and standard triangles. Finally, the *planting proposition* [Wue26, Prop. 3.1]: for reflexive Δ with unit edges, generic $X \in |-K|$ meets the toric curve $V(\sigma_F)$ of a two-dimensional face F in $\ell(F^\circ)$ reduced points in the open orbit, at each of which an implicit-function argument on the chart $U_{\sigma_F} \cong C(F) \times \mathbb{C}^*$ identifies the germ of X with $C(F)$. We use the standard normalization $\Delta^\circ = \{u : \langle u, v \rangle \geq -1 \ \forall v \in \Delta\}$, so that a facet of Δ corresponds to the vertex u of Δ° with $\langle u, \cdot \rangle = -1$ on the facet, and the dual edge of a two-dimensional face F is $F^\circ = \text{conv}\{u_1, u_2\} \subset \Delta^\circ$ for the two facets through F . (The prequel and the ancillary programs list facet data in the opposite normalization $\langle \cdot, \cdot \rangle = +1$; lattice lengths are unaffected.)

3. HOMOGENEOUS DEFORMATIONS ATTACHED TO TWO-DIMENSIONAL FACES

Petracci [Pet21] constructs, from a *deformation datum* for a cone, homogeneous deformations of affine toric varieties ([Pet21, Def. 3.1, Thm. 3.5], after Mavlyutov [Mav04]) and of polarized projective toric pairs ([Pet21, Thm. 6.1]). A datum for (N, σ) is a tuple $(Q, Q_0, \dots, Q_r, w)$, $w \in M$, with (i) $Q \subseteq \sigma$, (ii) $0 \notin Q$, (iii) $Q = Q_0 + \cdots + Q_r$, (iv) $Q_1, \dots, Q_r$ lattice polyhedra, (v) $\min_Q w \geq -1$, and (vi) every vertex of $\sigma \cap \{\langle w, \cdot \rangle = -1\}$ contained in $\mathbb{R}^+Q$.

Lemma 3.1 (codimension-two ∂ -datum). *Let $\Delta \subset \mathbb{Z}^4$ be reflexive, F a two-dimensional face with a Minkowski decomposition $F = Q_0 + \cdots + Q_r$ into lattice polytopes, and let $\tau = \text{Cone}(\Delta \times \{1\}) \subset \mathbb{Z}^5$, the cone of the polarized pair $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$. Let $u_1, u_2 \in \Delta^\circ$ be the vertices dual to the two facets of Δ containing F , so that the dual edge is $F^\circ = \text{conv}\{u_1, u_2\}$, and suppose $\ell(F^\circ) \geq 2$. Then for every interior lattice point v^* of F° , the tuple*

$$(F \times \{1\}, Q_0 \times \{1\}, Q_1, \dots, Q_r, w = (v^*, 0))$$

is a deformation datum for $(\mathbb{Z}^5, \tau)$ satisfying the lattice condition (iv), i.e. a ∂ -deformation datum in the sense of [Pet21, Def. 3.1].

Proof. Conditions (i)–(iv) are immediate ($Q = F \times \{1\}$ lies in τ at height one, and the summands may be translated so that $Q_1, \dots, Q_r$ are lattice polytopes and Q_0 carries the height). For (v) and (vi), evaluate w on the rays of τ , which are $(V, 1)$ for the vertices V of Δ : $\langle w, (V, 1) \rangle = \langle v^*, V \rangle \geq -1$ because $v^* \in \Delta^\circ$, with equality if and only if V lies on the face of Δ dual to the face of Δ° containing v^* in its relative interior. Since v^* is an interior lattice point of the edge F° , that dual face is exactly F : equality holds precisely at the vertices of F . Hence $\min_\tau w = -1$ is attained exactly on $\text{Cone}(F \times \{1\})$ — in particular $\min_Q w = -1$, which is (v) — the vertices of the slice $\tau \cap \{w = -1\}$ are the points $(V, 1)$, $V \in \text{vert}(F)$, and (vi) holds with $\mathbb{R}^+Q = \text{Cone}(F \times \{1\})$. $\square$

Remark 3.2. At the two endpoints of F° the slice grows to a full facet of Δ and condition (vi) would require a Minkowski decomposition of that three-dimensional facet; the interior points of the dual edge are exactly what makes the face-local datum possible. This is the source of the threshold $\ell(F^\circ) \geq 2$ in Theorem 1.1, and §5 shows the threshold is sharp.

By [Pet21, Thm. 6.1] the datum of Lemma 3.1 yields a flat deformation $(\mathcal{X}, \mathcal{D}) \rightarrow \mathbb{A}^r$ of the pair $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$, cut out by explicit binomials in the Cox coordinates of a projective toric variety built from the extended cone $\tilde{\tau} = \text{Cone}(\tau, Q_0 \times \{1\} - e_1 - \cdots - e_r, Q_1 + e_1, \dots, Q_r + e_r) \subset \mathbb{Z}^{5+r}$. We call this the *face family* of $(F, \sum Q_i, v^*)$.

Lemma 3.3 (chart identification). *Localized at the points of X over F , the face family induces on the germ $C(F)$ the Altmann–Cayley family of the decomposition $F = Q_0 + \cdots + Q_r$ (times a smooth trivial factor); in particular its general fibre near these points is the general fibre of the cascade lemma.*

Proof. Write N_F for the saturated sublattice of $\tilde{N} = \mathbb{Z}^5$ spanned by $\sigma_F = \text{Cone}(F \times \{1\})$, so that $N_F \cong \mathbb{Z}^3$ and $C(F) = \text{Spec } \mathbb{C}[\sigma_F^\vee \cap M_F]$ as in [Wue26, §2]; split $\tilde{N} = N_F \oplus \mathbb{Z}$ along the ray of the toric curve $V(\sigma_F)$, giving the chart $U_{\sigma_F} \cong C(F) \times \mathbb{C}^*$ of the planting proposition [Wue26, Prop. 3.1]. The auxiliary lattice of Petracci’s construction is $\tilde{N} \oplus \mathbb{Z}e_1 \oplus \cdots \oplus \mathbb{Z}e_r = N_F \oplus \mathbb{Z} \oplus \mathbb{Z}^r$.

Step 1: reduction to the affine germ. On the rays $(V, 1)$ of σ_F the functional $w = (v^*, 0)$ takes the value $\langle v^*, V \rangle = -1$ (proof of Lemma 3.1), so $w|_{N_F} = -u_F$, the *negative* of the Gorenstein degree of σ_F — which is exactly the degree in which the deformations of an isolated Gorenstein toric threefold singularity live [Alt95], cf. [Pet21, Rem. 3.7] — and $(F, Q_0, \dots, Q_r, -u_F)$ is a ∂ -deformation datum for (N_F, σ_F) . The extended cone $\tilde{\tau}$ meets $N_F \oplus \mathbb{Z}^r$ in the extended cone $\tilde{\sigma}_F$ of this restricted datum, and does not involve the $\mathbb{Z}$ -summand carrying the $\mathbb{C}^*$; hence, exactly as in the proof of [CHP24, Prop. 2.12], the restriction of the family $(\mathcal{X}, \mathcal{D}) \rightarrow \mathbb{A}^r$ to the star of the face $G \preceq \tilde{\tau}$ lying over σ_F is

$$(\text{Petracci’s affine family of } (N_F, \sigma_F, F = \sum Q_i)) \times \mathbb{C}^*.$$

It therefore suffices to identify the affine family, a deformation of $C(F)$ over $\mathbb{A}^r$.

Step 2: the Cayley cone. Because every Q_i is a unit segment or a standard triangle, the normal fans of the summands are transverse and each vertex of F has a unique expression $q_0 + q_1 + \cdots + q_r$ with $q_i \in \text{vert}(Q_i)$; correspondingly the rays of $\tilde{\sigma}_F$ over F are the vectors (q_0, h_0) and (q_j, h_j) ($j = 1, \dots, r$), where in the auxiliary coordinates (height, $e_1, \dots, e_r$) $\in \mathbb{Z}^{1+r}$ one has $h_0 = (1; -1, -1, \dots, -1)$ and $h_j = (0; \mathbf{e}_j)$. The $(r+1) \times (r+1)$ matrix with rows $h_0, h_1, \dots, h_r$ is $\begin{pmatrix} 1 & -\mathbf{1}^\top \\ 0 & I_r \end{pmatrix}$, which is unimodular; the inverse automorphism of $\mathbb{Z}^{1+r}$ sends h_i to the standard basis vector f_i , and hence carries the face $G = \text{Cone}(\bigcup_i Q_i \times \{h_i\})$ onto the standard Cayley cone $\text{Cone}(\bigcup_i Q_i \times \{f_i\})$ of the decomposition (for $r = 1$ this is the matrix $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$).

Step 3: the family is Altmann’s. Under the identification of Step 2, the projection $(e_1^*, \dots, e_r^*)$ that defines Petracci’s family [Pet21, Thm. 3.5] becomes the projection $(f_1^* - f_0^*, \dots, f_r^* - f_0^*)$ of the Cayley cone; this is precisely Altmann’s homogeneous one-parameter-per-summand deformation of $C(F)$ attached to $F = \sum_i Q_i$ [Alt95], whose realization on the Cayley cone and whose general fibre are recalled in [Wue26, Lemma 2.4]. Concretely, Petracci’s j -th binomial restricts on the chart of G to $\chi^{m_{j,1}} - \chi^{m_{j,0}} = t_j$, where $m_{j,0}, m_{j,1} \in \tilde{M}$ are the characters dual to the f_0 - and f_j -heights. For the type-(S) pentagon $F = T + s$, the Minkowski sum of a standard triangle T and a unit segment s , this is checked by machine

($r = 1$: the five vertices of F split uniquely, and $m_{1,0} = (0, 0, 1, 0, 0, 0)$, $m_{1,1} = m_{1,0} + e_1^*$; see the ancillary file `paper2_check.py`).

Step 4. By [Wue26, Lemma 2.4] the general fibre of this family is the cascade fibre of the decomposition: one singular point of type $C(Q_i)$ for each two-dimensional non-smooth summand. As every Q_i is a unit segment or standard triangle, all summand cones are smooth, so the general fibre is smooth near the points over F . This proves the lemma. $\square$

4. PROOF OF THEOREM 1.1

Theorem 4.1. *As stated in Theorem 1.1.*

We first record that the hypersurface follows the ambient family.

Lemma 4.2 (the anticanonical family). *Let $(\mathcal{X}, \mathcal{D}) \rightarrow (T, 0)$, $T \subseteq \mathbb{A}^r$ a neighbourhood of the origin, be the face family of Lemma 3.1, with central fibre $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$. Then for every section $f_0 \in H^0(\mathbb{P}_\Delta, -K)$ there is, after shrinking T , a section $f \in H^0(\mathcal{X}, \mathcal{O}_\mathcal{X}(\mathcal{D}))$ restricting to f_0 , and its zero locus $\mathcal{Y} \subset \mathcal{X}$ is flat over T with $\mathcal{Y}_0 = \{f_0 = 0\}$.*

Proof. By [Pet21, Thm. 6.1], $\mathcal{D}$ is a relative effective Cartier divisor on $\mathcal{X}$, flat over T , with central fibre $\partial\mathbb{P}_\Delta$. We first check that the nearby fibres $\mathcal{D}_t \subset \mathcal{X}_t$ are again *anticanonical*. Gorenstein-ness is open in a proper flat family, so after shrinking T all fibres $\mathcal{X}_t$ are Gorenstein and the relative dualizing sheaf $\omega_{\mathcal{X}/T}$ is a line bundle; set $\mathcal{M} = \omega_{\mathcal{X}/T}(\mathcal{D})$. On the central fibre $\mathcal{M}_0 = \omega_{\mathbb{P}_\Delta}(\partial\mathbb{P}_\Delta) \cong \mathcal{O}$, and $h^1(\mathbb{P}_\Delta, \mathcal{O}) = 0$; by cohomology and base change the trivializing section of $\mathcal{M}_0$ extends to a section of $\mathcal{M}$ after shrinking T , and by properness its zero locus, which misses $\mathcal{X}_0$, misses the nearby fibres. Hence $\omega_{\mathcal{X}_t}(\mathcal{D}_t) \cong \mathcal{O}_{\mathcal{X}_t}$: each $\mathcal{D}_t$ is anticanonical. Set $\mathcal{L} = \mathcal{O}_\mathcal{X}(\mathcal{D})$, a relative anticanonical bundle. On the central fibre, $-K_{\mathbb{P}_\Delta}$ is ample on the Gorenstein Fano toric fourfold $\mathbb{P}_\Delta$, so $h^i(\mathbb{P}_\Delta, -K) = 0$ for $i > 0$ by toric Kodaira vanishing. By semicontinuity the same vanishing holds on nearby fibres after shrinking T ; since $\chi(\mathcal{X}_t, \mathcal{L}_t)$ is constant in a flat family, $h^0(\mathcal{X}_t, \mathcal{L}_t)$ is constant, so by Grauert's criterion $\pi_*\mathcal{L}$ is locally free and its formation commutes with base change. Hence f_0 extends to a section f of $\mathcal{L}$. Its zero locus is a family of Cartier divisors in the fibres, and an algebraic family of divisors is flat [Har77, Ch. III, Example 9.8.5].

Finally, each fibre $\mathcal{Y}_t = \{f_t = 0\}$ is an anticanonical Cartier divisor in the Gorenstein $\mathcal{X}_t$, so adjunction gives $\omega_{\mathcal{Y}_t} \cong \mathcal{O}$, and $h^1(\mathcal{Y}_t, \mathcal{O}) = 0$ near $t = 0$ by semicontinuity from $h^1(\mathcal{Y}_0, \mathcal{O}) = 0$: the fibres are Calabi–Yau. $\square$

Proof of Theorem 4.1 (assembly). Let $(\mathcal{X}, \mathcal{D}) \rightarrow T$ be the face family and $\mathcal{Y} \rightarrow T$ the anticanonical family of Lemma 4.2 through a generic f_0 , so $X = \mathcal{Y}_0$.

The points over F smooth. Fix one of the $\ell(F^\circ)$ singular points p of X over F . By Lemma 3.3 the germ of the ambient family at p is the Cayley family of the decomposition times a smooth factor; since all summands Q_i are unit segments and standard triangles, the cascade lemma gives that the general fibre of that local family is *smooth* along the relevant stratum, and the transversality argument of the planting proposition, applied in the deformed chart (transversality is an open condition, and the intersection points stay in the corresponding open orbit for small t), shows that $\mathcal{Y}_t$ is smooth near p for generic small $t \in T$.

The unique-face case. Suppose F is the unique non-smooth two-dimensional face of Δ . Then X is smooth away from the $\ell(F^\circ)$ points over F (the planting proposition), so $\mathcal{Y}_0$ is smooth outside the former F -points; smoothness is open on the total space of the proper flat family $\mathcal{Y} \rightarrow T$, so after shrinking, $\mathcal{Y}_t$ is smooth away from a neighbourhood of those points for all small t , and smooth near them for generic t by the previous step. Restricting $\mathcal{Y}$ to

a generic line through $0 \in T$ gives a flat proper family over a disc with special fibre X and smooth Calabi–Yau general fibre (Lemma 4.2): X is smoothable. $\square$

Remark 4.3 (first-order persistence at the other faces). The face family does not disturb the other germs at first order. By [Pet21, Rem. 3.7] — stated there for the affine construction of [Pet21, Thm. 3.5]; the projective family carries the same parameter weights, as its trinomials do — the Kodaira–Spencer image of the family of Lemma 3.1 lies in the weight- w component of T^1 , with $w = (v^*, 0)$. For a face $F' \neq F$, the space $T_{C(F')}^1$ is concentrated in the degree $-u_{F'}$, the character taking the value -1 on all rays $(V', 1)$ of $\sigma_{F'}$ [Alt95]; but F' has a vertex V' not on F , where $\langle w, (V', 1) \rangle = \langle v^*, V' \rangle \geq 0 \neq -1$ (it is an integer ≥ -1 , and $= -1$ exactly on F by the proof of Lemma 3.1). So the induced first-order deformation of the germ over F' vanishes; likewise no positive multiple $|\alpha|w$ equals $-u_{F'}$, which by Rim’s equivariant structure on the miniversal deformation [Rim80] forces the higher-order Taylor coefficients of an equivariant classifying map into vanishing weight spaces as well. Upgrading this to constancy of the germs requires identifying the induced deformation of the affine chart along $V(\sigma_{F'})$ with an equivariant product family, as in Lemma 3.3 — routine for faces F' disjoint from F , where the chart of the extended cone over $\sigma_{F'}$ is untouched by the construction — and none of the results below use it, so we do not pursue it.

Theorem 4.4. *As stated in Theorem 1.2.*

Proof. (i) Let p lie over the face F_i and apply Theorem 4.1 to F_i : the resulting family $\mathcal{Y} \rightarrow T_i$ has generic fibre smooth near p . A semiuniversal deformation $\mathcal{S} \rightarrow (B, 0)$ of the compact complex space X exists by Grauert [Gra74]; choose a representative, proper over B , and shrink so that the relative singular locus $\Sigma = \text{Sing}(\mathcal{S}/B)$ is finite over B and meets the central fibre in $\text{Sing } X$. Then Σ decomposes as a disjoint union $\bigsqcup_q \Sigma_q$ over the singular points q of X , and each $D_q := \text{im}(\Sigma_q \rightarrow B)$ is a closed analytic subgerm (finite maps are proper). The family $\mathcal{Y} \rightarrow T_i$ is induced from $\mathcal{S}$ by a classifying map $\gamma_i : (T_i, 0) \rightarrow (B, 0)$; for generic t the fibre is smooth near p , so $\gamma_i(t) \notin D_p$: the subgerm D_p is proper.

(ii) After the shrinking above, a fibre $\mathcal{S}_b$ is smooth if and only if $b \notin D := \bigcup_q D_q$: away from the Σ_q the total space is smooth over B because the central fibre is (openness of smoothness). If $(B, 0)$ is irreducible, D — a finite union of proper closed analytic subgerms — is a proper closed analytic subgerm, so its complement meets every neighbourhood of 0. By the curve selection lemma for analytic germs (normalize B and slice by generic hyperplanes) there is a holomorphic arc $\varphi : (\mathbb{D}, 0) \rightarrow (B, 0)$ with $\varphi(\mathbb{D} \setminus 0) \not\subset D$; an irreducible arc not contained in the closed analytic set D meets it in a discrete set, so after shrinking the disc $\varphi(t) \notin D$ for all $t \neq 0$, and $\varphi^*\mathcal{S}$ is a smoothing of X .

(iii) If every F_i is a unit parallelogram, X is a nodal Calabi–Yau threefold and the MPCP resolution $\widehat{X} \rightarrow X$ is a projective *small* resolution (each square face has no interior or edge lattice points, so the maximal triangulation inserts only the diagonal wall). The $\ell(F_i^\circ) \geq 2$ nodes over one face lie on the single toric curve $V(\sigma_{F_i}) \cong \mathbb{P}^1$ and their exceptional curves are algebraically equivalent, hence homologous (§7); choosing integers $a_{i,1}, \dots, a_{i,\ell_i}$, all non-zero, with $\sum_j a_{i,j} = 0$ — possible exactly because $\ell_i \geq 2$ — and summing over the faces produces a relation $\sum a_k [C_k] = 0$ among *all* exceptional classes with every coefficient non-zero. By Friedman’s smoothing theorem for nodal Calabi–Yau threefolds with projective small resolution [Fri86] (see also [RT09]), X is smoothable. $\square$

Remark 4.5. The naive strategy for several faces — deform along the F_1 -family, then repeat for F_2 — does not make sense as it stands: the fibres of a face family are no longer toric hypersurfaces, so Lemma 3.1 cannot be applied to them. Nor do we know how to combine

the data for several faces into a single deformation: Petracci’s construction takes one datum, and the extended cones of different faces interfere. This is the same difficulty that leads Corti–Hacking–Petracci, one dimension down, to assemble their per-facet admissible Minkowski data not torically but through the semiuniversal deformation of a partial resolution with unobstructed deformations [CHP24]. Whether Theorem 4.4(ii) holds unconditionally — equivalently, whether the irreducibility hypothesis can be removed — is Question 8.2 below; note that by Gross [Gro97] the deformation space of a Calabi–Yau threefold with canonical singularities can be singular, so the hypothesis is not automatic.

4.1. The global cascade. Theorem 1.1 invoked the type (S) hypothesis only at the last step, to conclude that the local general fibre is smooth. The deformation datum and its chart identification are insensitive to the summand types, so the same family realizes globally the *cascade* of [Wue26, Lemma 2.4]: it moves the singular points over F to the germs of the two-dimensional non-smooth summands of a chosen Minkowski decomposition.

Proposition 4.6 (global cascade). *Let Δ be reflexive with unit edges whose unique non-smooth two-dimensional face F has a Minkowski decomposition $F = Q_0 + \cdots + Q_r$ into lattice polytopes and $\ell(F^\circ) \geq 2$. Then there is an explicit r -parameter deformation of the generic anticanonical X , given by binomial equations in the Cox coordinates, whose generic fibre is a Calabi–Yau threefold that has exactly $\ell(F^\circ)$ points of type $C(Q_i)$ for each two-dimensional non-smooth summand Q_i , and no other singular points. In particular, if F is deformable-but-non-smoothable (type (D) in the trichotomy of [Wue26]), so that its maximal decomposition has at least one rigid two-dimensional summand, then X deforms to a Calabi–Yau threefold in which each of the $\ell(F^\circ)$ points of type $C(F)$ is replaced by a rigid point $C(R_j)$ for each rigid summand R_j ; this deformed threefold is again non-smoothable.*

Proof. Lemma 3.1 builds the ∂ -datum from any Minkowski decomposition into lattice polytopes with $\ell(F^\circ) \geq 2$: its proof uses only that $F = \sum_i Q_i$ and $v^* \in \text{relint } F^\circ$, never the summand types. The one place Lemma 3.3 used type (S) is the unique splitting of each vertex of F through the summands; but a vertex of a Minkowski sum decomposes uniquely into vertices of the summands — its normal cone is the intersection of theirs — so that step, and the unimodular-height identification with the Altmann–Cayley cone, hold for an arbitrary decomposition. By the cascade lemma [Wue26, Lemma 2.4] the general fibre of the Altmann–Cayley family has exactly one singular point of type $C(Q_i)$ for each two-dimensional non-smooth summand Q_i , and is smooth elsewhere. Lemma 4.2 carries a generic anticanonical section along the family, with Calabi–Yau fibres; the transversality of the planting proposition, applied in the deformed chart, identifies the germs near the former F -points with the summand germs; and away from the former F -points the fibres are smooth, since X is (the face F being the unique non-smooth face) and smoothness is open on the total space of the proper flat family. When F is of type (D) the only two-dimensional non-smooth summands are the rigid R_j , so each $C(F)$ point becomes a collection of $C(R_j)$ points; since each $C(R_j)$ has no smoothing component (the trichotomy of [Wue26]), the deformed threefold admits no smoothing by the local-to-global lemma of [Wue26]. $\square$

Remark 4.7. The smallest instance is the def-only pentagon $Q_B = T + s$ of [Wue26] ($T = \frac{1}{3}(1, 1, 1)$, s a unit segment; its maximal decomposition is $T + s$ with the single rigid summand T), planted with $\ell(F^\circ) = 3$ as the unique non-smooth face of the polytope Δ_B of [Wue26]. The Calabi–Yau threefold X_B — whose three singular points are all deformable yet non-smoothable — then deforms, through an explicit binomial family, to a Calabi–Yau threefold with three ordinary $\frac{1}{3}(1, 1, 1)$ points, itself non-smoothable. (The Cayley bookkeeping of the

datum for this non-standard summand is verified in the ancillary file `cascade_check.py`.) Thus a single deformable-but-non-smoothable Calabi–Yau threefold lies in the closure of a family of quotient-singular ones: an explicit deformation between two non-smoothable compact Calabi–Yau threefolds, neither of them smoothable. It is the frozen analogue of a conifold transition — the local deformation genuinely moves, but only trades each non-smoothable germ for its rigid core, never reaching a smooth fibre.

5. SHARPNESS: A CALABI–YAU THREEFOLD WITH ONE RIGID NODE

Theorem 5.1. *As stated in Theorem 1.3. The MPCP resolution $\hat{X}_N$ has $(h^{1,1}, h^{2,1}) = (3, 103)$ and Euler number -200 .*

Proof. The verification that Δ_N is reflexive with unit edges, that its 21 two-dimensional faces comprise 20 standard triangles and the square with $\ell(F^\circ) = 1$, and the Hodge numbers, are finite computations performed by the programs of [Wue26]: the ten facet normals, each at level exactly 1, are

$$\begin{aligned} (0, 0, 1, 1), \quad (0, 0, 1, 0), \quad (-1, 0, 1, 1), \quad (-5, 2, 1, 1), \quad (0, -1, 0, 1), \\ (0, 2, 1, 1), \quad (0, 2, 1, -4), \quad (2, 0, -1, 1), \quad (2, 0, -1, -4), \quad (2, -5, -6, 1), \end{aligned}$$

with the dual edge of the square face spanned by the first two. By the planting proposition X_N has exactly one singular point, an ordinary double point p .

The square face has no interior lattice points and unit edges, so the MPCP subdivision of its cone inserts only the diagonal wall: the induced crepant resolution $\pi : \hat{X}_N \rightarrow X_N$ is *small*, with exceptional locus a single rational curve C over p , and it is projective because the MPCP subdivision is projective [Bat94]. If A is ample on $\hat{X}_N$ then $A \cdot C > 0$, so $[C] \neq 0$ in $H_2(\hat{X}_N; \mathbb{Q})$. By Friedman’s theorem [Fri86] (extended to all odd-dimensional nodal Calabi–Yau varieties in [RT09]), a nodal Calabi–Yau threefold with a projective small resolution is smoothable only if the exceptional classes satisfy a relation $\sum a_i [C_i] = 0$ in $H_2(\hat{X}_N; \mathbb{Q})$ with all $a_i \neq 0$; with a single node this forces $[C] = 0$. Hence X_N admits no smoothing. $\square$

Remark 5.2. No *polynomial* deformation (moving X_N inside $\mathbb{P}_{\Delta_N}$) can smooth the node — every Δ -regular member has the same singularity structure by the planting proposition — so Theorem 5.1 says the non-polynomial directions fail as well. Contrast the generic one-nodal quintic threefold, where the node smooths polynomially: there the small resolutions are non-Kähler and $[C]$ vanishes rationally. Δ_N manufactures the opposite situation inside the Kreuzer–Skarke list. Non-smoothable nodal Calabi–Yau threefolds as such are not new — the local-to-global failure for nodes is the classical content of [Fri86], and obstructed nodal configurations appear in Namikawa’s study of the local moduli of Calabi–Yau threefolds [Nam02] — what Δ_N adds is a one-node realization produced, and certified non-smoothable, by the combinatorics of a single reflexive polytope.

6. THE DEFECT BY GERM AT $\ell(F^\circ) = 1$: BEYOND FRIEDMAN AND GROSS

Proposition 6.1. *As stated in Proposition 1.4.*

Proof. The census of [Wue26] has exactly five classes with one interior point, the anticanonical cones over the five smooth toric del Pezzo surfaces; two of them, $\mathbb{P}^2$ and $\mathbb{F}_1$, are rigid, and the remaining three — the cones over $\mathbb{P}^1 \times \mathbb{P}^1$ (deg 8), dP_7 (deg 7) and dP_6 (deg 6) — are of type (S). So F is one of these three. An MPCP subdivision is a maximal *projective* crepant subdivision and is not unique; we choose one whose restriction to $\text{Cone}(F)$ is the star subdivision at the interior lattice point q of F : a pulling triangulation of the boundary that

pulls first at q is regular, hence projective, and pulling at q first forces every maximal cell over F to contain q [CLS11, Ch. 15]. The star subdivision of $\text{Cone}(F)$ at q is unimodular: after translating q to the origin, each triangle spanned by an edge of F and the origin has determinant ± 1 , a finite check for each of the three polygons (`paper2_check.py`). Hence this MPCP resolution is smooth over the point, with irreducible exceptional divisor the smooth toric del Pezzo surface S (the toric surface of the star of q) contracted to the point, and it is an isomorphism elsewhere; all other faces being smooth and all edges unit, $\hat{X}$ is smooth everywhere.

Now the negative statements. Let k be the number of vertices of F , so $\rho(S) = k - 2 \in \{2, 3, 4\}$. For each vertex ρ' of F the toric divisor $V(\rho') \cap \hat{X}$ of $\hat{X}$ restricts to S as the toric boundary curve of S corresponding to the edge (q, ρ') of the star subdivision; these boundary divisors generate $\text{Pic}(S)_{\mathbb{R}}$, so the intersection pairing of divisors of $\hat{X}$ with curves contained in S has rank $\rho(S)$, and the classes of the contracted curves span a face of dimension $\rho(S) \geq 2$ in $N_1(\hat{X})$: the relative Picard number of $\hat{X} \rightarrow X$ is $\rho(S) \geq 2$. By the relative cone theorem the morphism then factors as $\hat{X} \rightarrow Z \rightarrow X$ with both steps non-trivial (contract one extremal ray of $\overline{NE}(\hat{X}/X)$), so it is not primitive; and since every MPCP resolution differs from this one by flops, no choice is primitive. (Concretely: a (-1) -curve $C \subset S$ has $N_{C/\hat{X}} = N_{C/S} \oplus \omega_S|_C = \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and can be contracted separately.) Finally, the local class group of the germ $C(F)$ is the cokernel of $M_F \rightarrow \mathbb{Z}^k$ (one coordinate per ray of σ_F), of rank $k - 3 \geq 1$, so X is not $\mathbb{Q}$ -factorial at the point. Thus the hypotheses of both [Gro97, Thm. 5.8] (primitivity) and [Gro97, Thm. 4.3] ($\mathbb{Q}$ -factoriality of X) fail. $\square$

Remark 6.2. Gross's theorem [Gro97, Thm. 5.8] smooths exactly these germs when they are realized by *primitive* type II contractions — such realizations exist on non-toric Calabi–Yau threefolds, where the restriction $\text{Pic}(\hat{X}) \rightarrow \text{Pic}(S)$ can have rank one, and are constructed and smoothed in [Gro97, KK09] — and fails exactly for $E \cong \mathbb{P}^2, \mathbb{F}_1$, which are precisely the two $i = 1$ classes that are *rigid* in the trichotomy of [Wue26]: the local classification and the global defect see the same dichotomy. Proposition 6.1 says the Batyrev hypersurfaces never provide the primitive realization: at $\ell(F^\circ) = 1$ the toric mechanism of §3 produces no face datum, Friedman's obstruction has no known analogue for these germs, and Gross's theorems do not apply — the smoothability of these X is genuinely open (Question 8.1).

7. CONIFOLD CONFIGURATIONS AND THE BATYREV–KREUZER CENSUS

Suppose every non-smooth two-dimensional face of Δ is a unit parallelogram, so that X is a nodal Calabi–Yau threefold; this is the setting of Batyrev–Kreuzer [BK10]. The $\ell(F^\circ)$ nodes over one face F form a single algebraic family over $V(\sigma_F) \cong \mathbb{P}^1$, hence their exceptional curves in the MPCP resolution are homologous. Friedman's relation restricted to one face, $\sum_{i=1}^{\ell} a_i [C_F] = 0$ with all $a_i \neq 0$, is therefore satisfiable if and only if $\ell = \ell(F^\circ) \geq 2$ — reproducing the threshold of Theorem 1.1 from homology alone. A polytope in the BK population is thus smoothable whenever all its square faces have $\ell(F^\circ) \geq 2$ (Theorem 1.2(iii): sum the per-face relations and apply Friedman's smoothing theorem to the projective small MPCP resolution), while faces with $\ell(F^\circ) = 1$ require a *cross-face rescue*: the class $[C_F]$ must be a rational combination of the classes over other faces.

We have tabulated this over the *entire* Kreuzer–Skarke list. Our scan finds exactly 198,849 reflexive 4-polytopes in the BK population — every two-dimensional face a standard triangle or a unit parallelogram, and at least one parallelogram, so that X is nodal with at least one node (polytopes all of whose faces are standard triangles give smooth X and are excluded)

— in perfect agreement with the count of Batyrev–Kreuzer [BK10], an independent check on both computations. Of these, only 3,774 — that is, 1.90% — have *all* their square faces of dual length $\ell(F^\circ) \geq 2$, and are thus smoothable by the per-face argument alone; the remaining 195,075 each carry at least one $\ell(F^\circ) = 1$ square. The all- $\ell \geq 2$ fraction falls sharply with the vertex count, from over 90% at 6 vertices to well under 1% beyond 14. Batyrev–Kreuzer certify 30,241 of the population as smoothable by Namikawa’s criterion. We have not checked that every all- $\ell \geq 2$ polytope passes their criterion — both conditions are sufficient for smoothability, and neither is known to imply the other — but the comparison of counts already shows that at most 3,774 of the 30,241, about one in eight, can be explained by the per-face mechanism: at least 26,467 of Batyrev–Kreuzer’s certified smoothings involve an $\ell(F^\circ) = 1$ square whose exceptional class is rescued by a homology relation involving the classes over other faces. Understanding the combinatorial structure of these cross-face relations — the analogue, for the Calabi–Yau hypersurface, of the admissible Minkowski data of [CHP24] — is the central open problem left by this note.

8. OPEN QUESTIONS

Question 8.1. Is a Calabi–Yau threefold whose unique singular point has germ $C(F)$ with F of type (S) and $\ell(F^\circ) = 1$ ever smoothable? For $i = 1$ (the anticanonical cones over $\mathbb{P}^1 \times \mathbb{P}^1$, dP_7 , dP_6) Proposition 6.1 shows the toric resolution is never a primitive contraction and X is not $\mathbb{Q}$ -factorial, so Gross’s theorems do not apply; for $i \geq 2$ the crepant resolution contracts a reducible divisor. In neither case is a Friedman-type obstruction known, and no face datum exists.

Question 8.2. Is Theorem 1.2(ii) unconditional? Equivalently: can the face families of several type-(S) faces be realized simultaneously (a codimension-two analogue of the admissible Minkowski data of [CHP24]), or is the semiuniversal base of a Batyrev hypersurface with only type-(S) faces always irreducible? The fibres of a face family are not toric, so the construction cannot be iterated naively (Remark 4.5).

Question 8.3. Combined criterion: is X smoothable if and only if every non-smooth two-dimensional face is of type (S) and the exceptional classes over the $\ell(F^\circ) = 1$ faces admit a Friedman-type rescue? Together with [Wue26] this would decide smoothability for all unit-edge Batyrev hypersurfaces — a criterion fast enough to run over the full Kreuzer–Skarke list.

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